

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2025 Worked Solutions

January 2025 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2025

January 2025 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Arithmetic Sequences

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

- (a) the number of terms in the series,

(2)

- (b) the sum of the series.

(2)

Worked Solution - Question 1

1. Identify the arithmetic series

The first term is **2**, the common difference is **3**, and the last term is **254**.

2. Find the number of terms

Use $254 = 2 + (n - 1)3$. Then $252 = 3(n - 1)$, so $n - 1 = 84$ and $n = 85$.

3. Use the sum formula

$S_n = \frac{n}{2}(a + l)$, where a is the first term and l is the last term.

4. Find the sum

$$S = \frac{85}{2}(2 + 254) = 85 \times 128 = 10880.$$

Final answer

(a) 85.

(b) 10880.

Question 2

Binomial Expansion

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(2 - 5x)^8$$

giving each term in simplest form.

(4)

This expansion is to be used to find an approximation for 2.05^8

- (b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

Worked Solution - Question 2

1. Write the first four terms

$$(2 - 5x)^8 = 2^8 + 8(2^7)(-5x) + \binom{8}{2}(2^6)(-5x)^2 + \binom{8}{3}(2^5)(-5x)^3 + \dots$$

2. Simplify each term

The first four terms are $256 - 5120x + 44800x^2 - 224000x^3$.

3. Connect the expansion to 2.05

To use $(2 - 5x)^8$ for 2.05^8 , set $2 - 5x = 2.05$.

4. Find x

$$-5x = 0.05, \text{ so } x = -0.01.$$

Final answer

$$(a) 256 - 5120x + 44800x^2 - 224000x^3.$$

$$(b) x = -0.01.$$

Question 3

Applications of Differentiation

3.

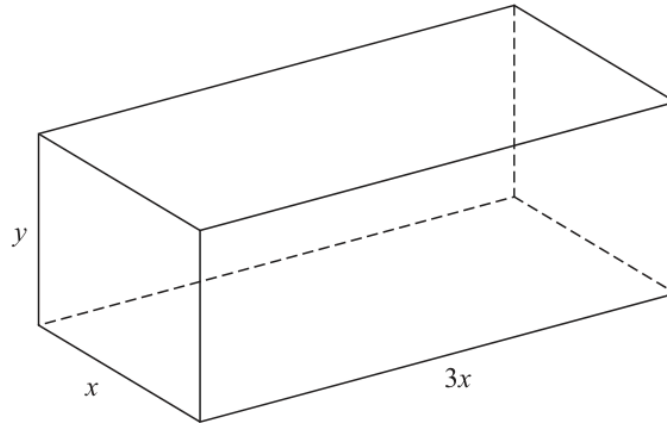


Figure 1

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by x metres.

The height of the container is y metres as shown in Figure 1.

Given that the capacity of the container is 120 m^3

(a) show that the area $A\text{ m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where P and Q are positive constants to be found.

(4)

(b) Use calculus to find the value of x for which A has a stationary value, giving your answer to 3 significant figures.

(4)

(c) Find $\frac{d^2A}{dx^2}$ and hence show that the value of x found in part (b) gives the minimum value of A .

(2)

Worked Solution - Question 3

Topic group

1. Use the volume

The volume is $(3x)(x)(y) = 3x^2y$. Since the capacity is 120, $3x^2y = 120$, so $y = \frac{40}{x^2}$.

2. Write the metal area

The open top means the metal area is the base plus four sides:

$$A = 3x^2 + 2(3xy) + 2(xy).$$

3. Substitute for y

$$A = 3x^2 + 8xy = 3x^2 + 8x \left(\frac{40}{x^2} \right) = 3x^2 + \frac{320}{x}. \text{ Hence } P = 3 \text{ and } Q = 320.$$

4. Differentiate

$$\frac{dA}{dx} = 6x - 320x^{-2} = 6x - \frac{320}{x^2}.$$

5. Use the stationary condition

$$\text{Set } \frac{dA}{dx} = 0: 6x = \frac{320}{x^2}, \text{ so } 6x^3 = 320.$$

6. Find x

$$x^3 = \frac{160}{3}, \text{ so } x = \sqrt[3]{\frac{160}{3}} = 3.764\dots \text{ To 3 significant figures, } x = 3.76.$$

7. Differentiate again

$$\frac{d^2A}{dx^2} = 6 + 640x^{-3} = 6 + \frac{640}{x^3}.$$

8. Show the minimum

For $x > 0$, $6 + \frac{640}{x^3} > 0$, so the stationary value found in part (b) is a minimum.

Final answer

(a) $A = 3x^2 + \frac{320}{x}$, so $P = 3$ and $Q = 320$.

(b) $x = 3.76$.

(c) $\frac{d^2A}{dx^2} = 6 + \frac{640}{x^3}$, minimum.

Question 4

Geometric Sequences

4.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(a) the common ratio,

(4)

(b) the first term.

(2)

Worked Solution - Question 4

1. Write the two sums

For first term a and common ratio r , $S_{\infty} = \frac{a}{1-r} = 75$ and

$$S_3 = \frac{a(1-r^3)}{1-r} = 70.2.$$

2. Divide the equations

$$\frac{S_3}{S_{\infty}} = 1 - r^3, \text{ so } \frac{70.2}{75} = 1 - r^3.$$

3. Find the ratio

$$0.936 = 1 - r^3, \text{ so } r^3 = 0.064 \text{ and } r = 0.4.$$

4. Find the first term

$$\text{Use } \frac{a}{1-r} = 75. \text{ Hence } a = 75(0.6) = 45.$$

Final answer

(a) $r = 0.4$.

(b) 45.

Question 5

Polynomials

5.

$$f(x) = 3x^3 + ax^2 - 10x + b$$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,

(b) find the value of a and the value of b .

(4)

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

Worked Solution - Question 5

1. Use the factor theorem

Since $3x - 4$ is a factor, $x = \frac{4}{3}$ is a root and $f\left(\frac{4}{3}\right) = 0$.

2. Show the given equation

$$3\left(\frac{4}{3}\right)^3 + a\left(\frac{4}{3}\right)^2 - 10\left(\frac{4}{3}\right) + b = 0, \text{ so } \frac{64}{9} + \frac{16a}{9} - \frac{120}{9} + b = 0.$$

Multiplying by 9 gives $16a + 9b = 56$.

3. Use the remainder condition

When dividing by $x - 2$, the remainder is $f(2)$. Given that the remainder is b , $f(2) = b$.

4. Find a

$$f(2) = 24 + 4a - 20 + b = 4 + 4a + b. \text{ Thus } 4 + 4a + b = b, \text{ so } a = -1.$$

5. Find b

Substitute $a = -1$ into $16a + 9b = 56$: $-16 + 9b = 56$, so $b = 8$.

6. Write f of x

$$f(x) = 3x^3 - x^2 - 10x + 8.$$

7. Find another factor

$$f(1) = 3 - 1 - 10 + 8 = 0, \text{ so } (x - 1) \text{ is a factor.}$$

8. Fully factorise

Dividing by $(x - 1)$ gives $3x^2 + 2x - 8 = (3x - 4)(x + 2)$. Hence

$$f(x) = (x - 1)(x + 2)(3x - 4).$$

Final answer

(a) Proven.

(b) $a = -1$, $b = 8$.

(c) $f(x) = (x - 1)(x + 2)(3x - 4)$.

Question 6

6.

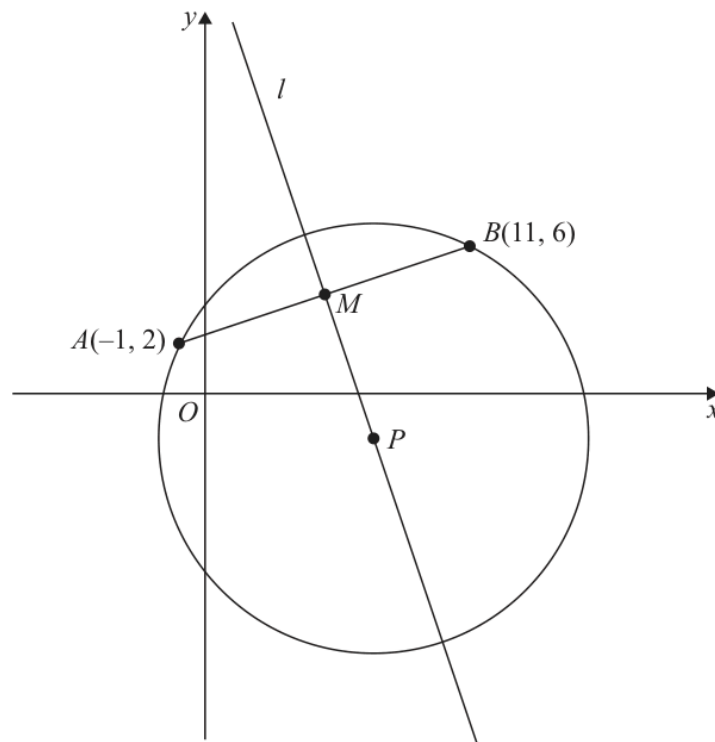


Figure 2

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants.

(4)

Given that P has coordinates $(7, k)$, where k is a constant,

- (b) find the value of k ,

(1)

- (c) find an equation for the circle.

(3)

Worked Solution - Question 6

1. Find the midpoint

For $A(-1, 2)$ and $B(11, 6)$, the midpoint is $M\left(\frac{-1+11}{2}, \frac{2+6}{2}\right) = (5, 4)$.

2. Find the gradient of AB

The gradient of AB is $\frac{6-2}{11-(-1)} = \frac{4}{12} = \frac{1}{3}$.

3. Use the perpendicular bisector

The centre lies on the perpendicular bisector of the chord AB , so line l has gradient -3 and passes through $M(5, 4)$.

4. Find line l

$y - 4 = -3(x - 5)$, so $y = -3x + 19$.

5. Use the centre coordinates

Since $P = (7, k)$ lies on l , $k = -3(7) + 19 = -2$.

6. Find the radius squared

The centre is $(7, -2)$. Using point A ,
 $r^2 = (7 - (-1))^2 + (-2 - 2)^2 = 8^2 + (-4)^2 = 80$.

7. Write the circle equation

Therefore the circle is $(x - 7)^2 + (y + 2)^2 = 80$.

Final answer

(a) $y = -3x + 19$.

(b) $k = -2$.

(c) $(x - 7)^2 + (y + 2)^2 = 80$.

Question 7

Laws of Logarithms

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

- (ii) Find the value of a such that

$$2\log_5(5 - a) - \log_5(a + 25) = 1$$

(5)

Worked Solution - Question 7

1. Use the trapezium rule

The width is 1, so

$$\int_{-1}^4 \log_{10}(x+5) dx \approx \frac{1}{2} \{ \log_{10} 4 + \log_{10} 9 + 2(\log_{10} 5 + \log_{10} 6 + \log_{10} 7 + \log_{10} 8) \}$$

2. Combine the logarithms

This is $\log_{10} (\sqrt{4 \times 9} \times 5 \times 6 \times 7 \times 8)$.

3. Find k

$$\sqrt{4 \times 9} \times 5 \times 6 \times 7 \times 8 = 6 \times 1680 = 10080, \text{ so } k = 10080.$$

4. State the domain

The logarithms require $5 - a > 0$ and $a + 25 > 0$, so $-25 < a < 5$.

5. Combine the logarithms

$$2 \log_5(5 - a) - \log_5(a + 25) = \log_5 \left(\frac{(5 - a)^2}{a + 25} \right).$$

6. Use one as a logarithm

$$1 = \log_5 5, \text{ so } \frac{(5 - a)^2}{a + 25} = 5.$$

7. Form the quadratic

$$(5 - a)^2 = 5(a + 25), \text{ so } a^2 - 10a + 25 = 5a + 125. \text{ Hence } a^2 - 15a - 100 = 0.$$

8. Solve and check

$$(a - 20)(a + 5) = 0, \text{ so } a = 20 \text{ or } a = -5. \text{ Only } a = -5 \text{ is in the domain, so } a = -5.$$

Final answer

(i) $k = 10080$.

(ii) $a = -5$.

Question 8

Proof

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

- (ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

Worked Solution - Question 8

Topic group

1. Give a counterexample

Let $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$. Both are irrational and $x \neq y$, but $xy = 1$, which is rational.

2. Represent an odd integer

Let $n = 2q + 1$, where q is an integer.

3. Substitute into the expression

$$n^3 + 3n + 2 = (2q + 1)^3 + 3(2q + 1) + 2.$$

4. Expand and factor

$$n^3 + 3n + 2 = 8q^3 + 12q^2 + 12q + 6 = 2(4q^3 + 6q^2 + 6q + 3).$$

5. Show it is even

The expression has a factor of **2**, so it is even.

6. Show it is not a multiple of four

The bracket $4q^3 + 6q^2 + 6q + 3$ is odd, because the first three terms are even and **3** is odd. Therefore the expression is **2** times an odd integer, so it is not divisible by **4**.

Final answer

(i) For example, $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$ give $xy = 1$.

(ii) Proven.

Question 9

Integration

9.

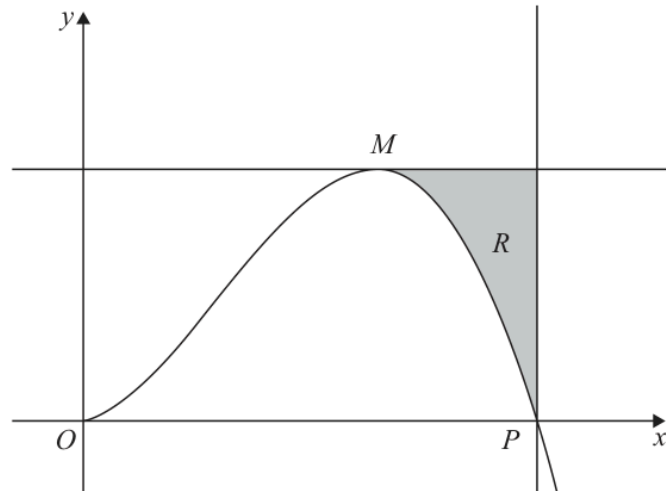


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

(c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

Worked Solution - Question 9

1. Rewrite the curve

$$y = \frac{9x^2(5 - \sqrt{x})}{5} = 9x^2 - \frac{9}{5}x^{5/2}.$$

2. Differentiate

$$\frac{dy}{dx} = 18x - \frac{9}{2}x^{3/2} = \frac{9}{2}x(4 - \sqrt{x}).$$

3. Find the turning point x coordinate

For the turning point shown away from the origin, set $4 - \sqrt{x} = 0$. Hence $x = 16$.

4. Find the y coordinate

$$\text{At } x = 16, y = \frac{9(16)^2(5 - 4)}{5} = \frac{2304}{5}.$$

5. State M

Therefore $M\left(16, \frac{2304}{5}\right)$.

6. Find the positive x intercept

For the positive intercept, $\frac{9x^2(5 - \sqrt{x})}{5} = 0$ gives $5 - \sqrt{x} = 0$, so $x = 25$.

7. Set up the area

The top boundary is $y = \frac{2304}{5}$ and the curve is below it from $x = 16$ to $x = 25$.

$$\text{So } R = \int_{16}^{25} \left(\frac{2304}{5} - 9x^2 + \frac{9}{5}x^{5/2} \right) dx.$$

8. Integrate

$$R = \left[\frac{2304}{5}x - 3x^3 + \frac{18}{35}x^{7/2} \right]_{16}^{25}.$$

9. Evaluate

$$R = \frac{9189}{7} = 1312.714\dots, \text{ so the area is } \mathbf{1312.7} \text{ to one decimal place.}$$

Final answer

(a) $M\left(16, \frac{2304}{5}\right).$

(b) $x_P = 25.$

(c) $1312.7.$

Question 10

Trigonometric Equations

10.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \quad \theta \neq \frac{n\pi}{2} \quad (4)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)

Worked Solution - Question 10

1. Expand the left side

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) = 3 \sin \theta + \frac{2 \cos \theta}{\tan \theta}.$$

2. Rewrite the second term

$$\text{Since } \tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \frac{2 \cos \theta}{\tan \theta} = \frac{2 \cos^2 \theta}{\sin \theta}.$$

3. Use the identity

$$3 \sin \theta + \frac{2 \cos^2 \theta}{\sin \theta} = 3 \sin \theta + \frac{2(1 - \sin^2 \theta)}{\sin \theta}.$$

4. Simplify

$$3 \sin \theta + \frac{2}{\sin \theta} - 2 \sin \theta = \sin \theta + \frac{2}{\sin \theta}, \text{ as required.}$$

5. Use part a

$$\text{The equation becomes } \sin x + \frac{2}{\sin x} = 4 \sin x - 5.$$

6. Form a quadratic in $\sin x$

$$\begin{aligned} \text{Multiplying by } \sin x \text{ gives } \sin^2 x + 2 &= 4 \sin^2 x - 5 \sin x, \text{ so} \\ 3 \sin^2 x - 5 \sin x - 2 &= 0. \end{aligned}$$

7. Solve for $\sin x$

$$(3 \sin x + 1)(\sin x - 2) = 0. \text{ Since } \sin x = 2 \text{ is impossible, } \sin x = -\frac{1}{3}.$$

8. Find x in the interval

For $0 < x < 2\pi$, $\sin x = -\frac{1}{3}$ gives $x = \pi + \sin^{-1}\left(\frac{1}{3}\right)$ or
 $x = 2\pi - \sin^{-1}\left(\frac{1}{3}\right)$.

9. Round the answers

Thus $x = 3.481\dots$ or $x = 5.943\dots$, so to 3 significant figures $x = 3.48$ or
 $x = 5.94$.

Final answer

(a) Proven.

(b) $x = 3.48, 5.94$.